

$$v_{\pi}(s) = q_{\pi}(s, \pi(s))$$

$$q_{\pi}(s, a) \doteq \mathbb{E}_{\pi} [G_t | S_t = s, A_t = a] = \sum_{s'} p(s'|s, a) [r(s, a) + \gamma v_{\pi}(s')] = \sum_{s'} p(s'|s, \pi(s)) [r(s, \pi(s)) + \gamma v_{\pi}(s)]$$

$$q_{\pi}(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_{\pi}(s')$$

$$v_{\pi}(s) \doteq \sum_{s'} \pi(a|s) \sum_{s''} p(s'', s'|s, a) [r + \gamma v_{\pi}(s'')] \quad \forall s \in S$$

$$q'_{\pi}(s, a) \doteq \sum_{s'} p(s'|s, a) [r(s, a) + \gamma q_{\pi}(s', a)]$$

$$q_{\pi}(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) q_{\pi}(s', \pi(s'))$$

$$q_{\pi}(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \sum_{a'} \pi(a'|s') q_{\pi}(s', a')$$

$$\begin{cases} v_{\pi}(\text{low}) - (0.9\theta v_{\pi}(\text{low}) - 0.9 v_{\pi}(\text{high}) - \theta(0.9) v_{\pi}(\text{high})) = 3\theta \\ v_{\pi}(\text{high}) - (0.63 v_{\pi}(\text{high}) - 0.27 v_{\pi}(\text{low})) = 10 \end{cases}$$

cont.

$$v_{\pi}(\text{low}) (1 - 0.9\theta) + v_{\pi}(\text{high}) (-0.9 - 0.9\theta) = 3\theta$$

$$v_{\pi}(\text{high}) (1 - 0.63) - 0.27 v_{\pi}(\text{low}) = 10$$

0.37

$$\begin{bmatrix} -0.9 - 0.9\theta & 1 - 0.9\theta & 3\theta \\ 0.37 & -0.27 & 10 \end{bmatrix}$$

* ideally want θ

to lead to $v_{\pi}(\text{high})$

being higher than $v_{\pi}(\text{low})$

and to lead to the robot working

to choose recharge over work

in the low state to avoid

running out of power

Ex 2

1) 4 - Rooms domain

$A = \{ \text{up, down, left, right} \}$ action space is the space containing all actions the agent is capable of

S^* , state space, collection of all possible states agent can be in (including terminal (10,10)).
 $10 \times 10 \rightarrow$ can't enter the wall spaces, not in state space

$\hookrightarrow 10 \times 10 \leftarrow 17$ wall spaces $\Rightarrow 100 - 17 \Rightarrow 83$ spaces

⑥ dynamics function $p(s', r | s, a)$ $S \times R \times S \times A \rightarrow [\emptyset, 1]$
 dynamics function $\rightarrow \emptyset$ when $\hookrightarrow 83$ states $\hookrightarrow 4$ actions

$r(s, a) = \emptyset$ everywhere $s' \in S$ and $r \in R$ are not possible

but $s' = (10, 10) \forall k \in \mathbb{N}$

$r(s', a) = 1$

approx $S \rightarrow s' \times 4$, $r = 0/1$, you can "roughly" transfer to 4 states from any state (edges, walls, change that), can choose 4 actions, each action can transfer to 3 (or less) states, rewards are $\emptyset$ everywhere,

- except edges, next to wall
 $\approx \begin{matrix} s & a & s' & r \\ 83 & 4 & 3 & 1 \end{matrix} \approx 996 \text{ approx.}$
 4 actions for each state transition
 $S = 83 \approx 3 \times 3$ next states

⑦ failure $\rightarrow$ failure $\rightarrow$ failure $\rightarrow$ failure
 $\hookrightarrow$ forces to cut $\rightarrow$ return at each timestep would be related to $-\gamma^{K-1}$ where K is # steps before failure (continuing)

failure $\rightarrow$ failure $\rightarrow$ failure $\rightarrow$ failure

γ discount, -1 for failure, $\emptyset$ else

$\hookrightarrow$ forces to cut $\rightarrow$ return at each timestep would be related to $-\gamma^{K-1}$ where K is # steps before failure (continuing)

return $G_t = R_{t+1} + \gamma G_{t+1}$

$\hookrightarrow$ no rewards except failure "reward", so for each timestep in episode there is no reward until the episode ends, with failure.

episode

$G_t = R_{t+1} + R_{t+2} + \dots + R_T \rightarrow$ all $\emptyset$ except failure episode ends

w/ discount $\rightarrow G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^k R_{t+k+1}$
 $\hookrightarrow$ all R minus R_T are $\emptyset$, R_T is -1 $\rightarrow \gamma^k R_{t+k+1} = \gamma^k R_{t+1} = \gamma^k$

$(-\gamma^{T-t-1} = G_t)$

continuing $G_t = R_{t+1} + \gamma R_{t+2} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

2) ①

reset to an episode; +1 for escaping, 0 else; episodic; $\gamma=1$

goal maximize reward

- With no penalty for being in the maze, the agent has no incentive to complete the maze.

- With $\gamma=1$ the Σ sign: sum goes off to ∞ .

infinity, and return will be 1 (since completion

regardless of episode length will result in a reward of 1 at termination.)

episode

episode

3) $\gamma = 0.5$; $R_1 = -1, R_2 = 2, R_3 = 6, R_4 = 3, R_5 = 2, T = 5$

$G_5 = R_{t+1} + \gamma G_{t+1} \quad t=5=T \rightarrow G_5 = R_6 + \gamma G_6 \quad G_5 = 0$

$G_5 = 0$

$G_4 = R_5 + \gamma G_5 = 2 + \gamma(0) = 2 = G_4$

$G_3 = R_4 + \gamma G_4 = 3 + (0.5)2 = 4 = G_3$

$G_2 = R_3 + \gamma G_3 = 6 + (0.5)4 = 8 = G_2$

$G_1 = R_2 + \gamma G_2 = 2 + (0.5)8 = 6 = G_1$

$G_0 = R_1 + \gamma G_1 = -1 + (0.5)6 = 2 = G_0$

④ $\gamma = 0.9$; $R_1 = 2, R_{\infty} = 7$

$G = R_1 + \gamma R_2 + \gamma^2 R_3 + \dots$

$G = 2 + \gamma 7 + \gamma^2 7 + \gamma^3 7 + \gamma^4 7 + \dots$

$G_1 = R_2 + \gamma G_2$

$\hookrightarrow \sum_{k=0}^{\infty} \gamma^k 7 \rightarrow 7 \sum_{k=0}^{\infty} \gamma^k = 7(1-\gamma) = 70 = G_1$

$G_0 = R_1 + \gamma G_1 \Rightarrow 2 + (0.9)(70) = 65 = G_0$

5) +1 for terminal state, -R for edge of world, 0 else

Eqn $G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

$v_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t | S_t = s]$

$= \mathbb{E}_{\pi} [R_{t+1} + \gamma G_{t+1} | S_t = s] = \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r | s, a)$

• getting C in front of rewards

$= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_{\pi}(s')]$

$G_t = C R_{t+1} + \gamma C R_{t+2} + \gamma^2 C R_{t+3} + \dots \Rightarrow C \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

↳ all terms have C, pull out

maximize $\downarrow$ no C. $\downarrow$ const.

$G_t \doteq G_t + \sum_{k=0}^{\infty} \gamma^k C$

↳ rewards

$v_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t^x | S_t = s]$

$v_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t + \frac{C}{1-\gamma} | S_t = s]$

⑥ $\sum_{k=0}^{\infty} \gamma^k = \frac{a(1-r^{n+1})}{1-r}$

$G_t = R_{t+1} + R_{t+2} + \dots + R_T$

→ negative rewards to push agent to goal

↳ positive rewards encourage agent to remain in maze and not terminate the task.

↳ small negative rewards encourage more time to complete episode, longer episodes.

6) $v_{\pi}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma v_{\pi}(s')] \quad \forall s \in S$
 • deterministic movement $\gamma = 0.9$

	2.3	
0.7	0.7	0.4
	-0.4	

$$v(\text{center}) = \frac{1}{4}(\text{UP})(\checkmark + \gamma(\text{center} + \text{UP})) + \frac{1}{4}(\text{Down})(\checkmark + \gamma(\text{center} + \text{Down}))$$

$$+ \frac{1}{4}(\text{LEFT})(\checkmark + \gamma(\text{center} + \text{left})) + \frac{1}{4}(\text{RIGHT})(\checkmark + \gamma(\text{center} + \text{right}))$$

$$v(\text{center}) = (0.25)(0.9)(2.3) + (0.25)(0.9)(-0.4) + (0.25)(0.9)(0.7) + (0.25)(0.9)(0.4)$$

$$(0.5175) + (-0.09) + (0.1575) + (0.09) = 0.675$$

	19.8	
19.8	17.8	16.0
	16.0	

② $v_{\pi} = \pi$

$$= [(\gamma 19.8 + \checkmark) + (\gamma 19.8 + \checkmark)] / 2 = 17.82 = 17.8$$

optimal policy shows 2 equiprobable actions at state

$$[2 \times ?] / 2 = 17.8$$

4)

5) (1)



Starts in A, L/R terminate

$\gamma = 1$

guess: $L = \emptyset, R = 1, A = \frac{1}{2}$

$$R = (1)(1 + \gamma \emptyset) = 1$$

$$L = (1)(\emptyset + \gamma \emptyset) = \emptyset$$

$$A = \frac{1}{2}(\emptyset + \gamma R) + \frac{1}{2}(\emptyset + \gamma L)$$

$$A = \frac{1}{2}(R) + \frac{1}{2}(L) = \frac{1}{2}(1) + \emptyset$$

$$A = \frac{1}{2}$$

guess: $L = \emptyset, R = 1, A = \frac{1}{2}, B = \frac{1}{6}, C = \frac{1}{6}, D = \frac{1}{6}, E = \frac{1}{6}$

$$L = (1)(\emptyset + \gamma \emptyset) = 1$$

$$R = (1)(\emptyset + \gamma \emptyset) = \emptyset$$

$$A = \frac{1}{2}(\emptyset + \gamma L) + \frac{1}{2}(\emptyset + \gamma B)$$

$$B = \frac{1}{2}(\emptyset + \gamma A) + \frac{1}{2}(\emptyset + \gamma C)$$

$$C = \frac{1}{2}(\emptyset + \gamma B) + \frac{1}{2}(\emptyset + \gamma D)$$

$$D = \frac{1}{2}(\emptyset + \gamma C) + \frac{1}{2}(\emptyset + \gamma E)$$

$$E = \frac{1}{2}(\emptyset + \gamma D) + \frac{1}{2}(\emptyset + \gamma R)$$

A	B	C	D	E		ref	A
1	$-\frac{1}{2}$	0	0	0	0	1	0
$-\frac{1}{2}$	1	$-\frac{1}{2}$	0	0	0	0	$\frac{1}{3}$
0	$-\frac{1}{2}$	1	$-\frac{1}{2}$	0	0	0	$\frac{1}{2}$
0	0	$-\frac{1}{2}$	1	$-\frac{1}{2}$	0	0	$\frac{2}{3}$
0	0	0	$-\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{5}{6}$

$L=0, A=\frac{1}{6}, B=\frac{1}{3}, C=\frac{1}{2}, D=\frac{2}{3}, E=\frac{5}{6}, R=1$

$\hookrightarrow$ 6 states $(\frac{1}{6})$ $(\frac{1}{3})$ $(\frac{1}{2})$ $(\frac{2}{3})$ $(\frac{5}{6})$

$E = \frac{1}{2}(\phi + \gamma P) + \frac{1}{2}(\phi + \gamma R)$

$L = 0, A = \frac{1}{6}, B = \frac{1}{3}, C = \frac{1}{2}, D = \frac{2}{3}, E = \frac{5}{6}, R = 1$
 6 states ($\frac{2}{6}$) ($\frac{4}{6}$) ($\frac{6}{6}$)

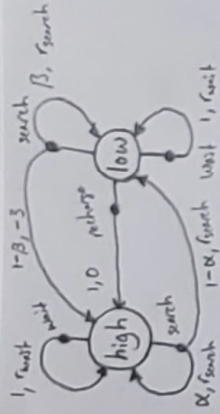
From terminal state w/ + reward to terminal state + 1 reward

for 0 to n states (S^+)

$V_n(s) = \frac{n}{(n-1)}$, for $\emptyset$ to $(n-1)$ where $n = \text{size of } S^+$

$$\frac{0}{6}, \frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}, \frac{6}{6}$$

0 1 2 3 4 5 6



8) a

arbitrary $\pi(a|s)$, discount factor γ , domain parameters $\alpha, \beta, r_{\text{search}}, r_{\text{wait}}$
 Bellman eqn $v_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) [r + \gamma v_{\pi}(s')]$

$$v_{\pi}(\text{high}) = \pi(\text{search}|\text{high}) [\alpha \pi(\text{search}|\text{high}) + (1-\alpha) \pi(\text{wait}|\text{high})] + \alpha [\pi(\text{search} + \gamma v_{\pi}(\text{high})) + (1-\alpha) [\pi(\text{search} + \gamma v_{\pi}(\text{low}))]]$$

$$v_{\pi}(\text{low}) = \pi(\text{search}|\text{low}) [\beta (\pi(\text{search} + \gamma v_{\pi}(\text{low})) + (1-\beta) (-3 + \gamma v_{\pi}(\text{high}))) + \pi(\text{wait}|\text{low})] + (1-\beta) (-3 + \gamma v_{\pi}(\text{high}))$$

⑤ $\alpha = 0.7, \beta = 0.6, \gamma = 0.9, r_{\text{search}} = 10, r_{\text{wait}} = 3$
 $\pi(\text{search}|\text{high}) = 1, \pi(\text{wait}|\text{low}) = 0.5, \pi(\text{recharge}|\text{low}) = 0.5$
 $v_{\pi}(\text{wait}|\text{high}) = \emptyset, \pi(\text{search}|\text{low}) = \emptyset, v_{\pi}(\text{high}), v_{\pi}(\text{low}) = ?$

$$v_{\pi}(\text{high}) = (1) [(0.7)(10 + (0.9)v_{\pi}(\text{high})) + (0.3)(10 + (0.9)v_{\pi}(\text{low}))]$$

$$v_{\pi}(\text{low}) = (0.5) [(1)(3 + (0.9)v_{\pi}(\text{low}))] + (0.5) [(1)(\emptyset + (0.9)v_{\pi}(\text{high}))]$$

$$\begin{cases} v_{\pi}(\text{high}) = 7 + 0.63 v_{\pi}(\text{high}) + 3 + 0.27 v_{\pi}(\text{low}) \\ v_{\pi}(\text{low}) = 1.5 + 0.45 v_{\pi}(\text{low}) + \emptyset + 0.45 v_{\pi}(\text{high}) \end{cases}$$

$$\begin{cases} v_{\pi}(\text{high}) - 0.63 v_{\pi}(\text{high}) - 0.27 v_{\pi}(\text{low}) = 10 \\ v_{\pi}(\text{low}) - 0.45 v_{\pi}(\text{low}) - 0.45 v_{\pi}(\text{high}) = 1.5 \end{cases}$$

$$\begin{cases} 0.37 v_{\pi}(\text{high}) - 0.27 v_{\pi}(\text{low}) = 10 \\ 0.45 v_{\pi}(\text{low}) - 0.45 v_{\pi}(\text{high}) = 1.5 \end{cases}$$

$$\begin{cases} v_{\pi}(\text{high}) = 109 \\ v_{\pi}(\text{low}) = 112.3 \end{cases}$$

⑥ $\pi(\text{wait}|\text{low}) = \emptyset, \pi(\text{recharge}|\text{low}) = 1-\emptyset \Rightarrow \pi(\text{search}|\text{low}) = \emptyset$
 $v_{\pi}(\text{low}) = \pi(\text{search}|\text{low}) [\beta (\pi(\text{search} + \gamma v_{\pi}(\text{low})) + (1-\beta) (-3 + \gamma v_{\pi}(\text{high}))) + \pi(\text{wait}|\text{low})] + (1-\beta) (-3 + \gamma v_{\pi}(\text{high}))$
 $+ \pi(\text{recharge}|\text{low}) [(1)(\emptyset + \gamma v_{\pi}(\text{low}))]$
 $+ \pi(\text{search}|\text{low}) [(1)(\emptyset + \gamma v_{\pi}(\text{high}))]$

$$v_{\pi}(\text{low}) = (\emptyset) [(1)(3 + (0.9)v_{\pi}(\text{low}))] + (1-\emptyset) [(1)(\emptyset + (0.9)v_{\pi}(\text{high}))]$$

$$v_{\pi}(\text{low}) = 3\emptyset + (0.9)\emptyset v_{\pi}(\text{low}) + (0.9)v_{\pi}(\text{high}) - \emptyset(0.9)v_{\pi}(\text{high})$$

$$v_{\pi}(\text{high}) = 7 + 0.63 v_{\pi}(\text{high}) + 3 + 0.27 v_{\pi}(\text{low})$$